

Module 01: Systems in Embodied

Learning Objectives

1. Define **Systems** within the context of Embodied.
2. Analyze how **Systems** interacts with other components of the Active Inference framework.
3. Apply specific constraints of Embodied to the formal definition of **Systems**.

Introduction

This module explores **Systems**. In the **Embodied** curriculum, we approach this topic with a focus on specific applications and theoretical depth appropriate for the audience. **Systems** is a critical component of the 8-part Active Inference spine, bridging the gap between Planning and Agents.

Key Concepts

1. Systems as a Markov Blanket Boundary

How does **Systems** define the boundary between the agent and the environment?

2. Generative Models of Systems

What parameters involved in **Systems** must be optimized to minimize variational free energy?

3. Active Inference Dynamics

How does the process of **Systems** drive the perception-action loop?

Applications

In Embodied, we see **Systems** manifest in: * **Specific Example 1**: When I run along a rocky trail, my musculoskeletal, vestibular, and visual systems operate as a tightly coupled

locomotor system -- each footfall sends proprioceptive signals upward while the vestibular apparatus predicts the next head position and the visual system scans for obstacles ahead; this whole-body movement system maintains its Markov blanket dynamically, continuously minimizing the free energy of postural instability by coordinating dozens of joints within milliseconds. * **Specific Example 2:** A dancer performing a complex choreography functions as an integrated neuro-musculo-skeletal system where respiratory rhythm, spinal undulation, limb trajectory, and spatial orientation are all coupled through shared generative model predictions; the experience of being "in the flow" of movement is the subjective signature of systemic free energy minimization -- the entire body-as-system operating in a regime of minimal prediction error across all sensorimotor channels simultaneously.

Conclusion

Understanding Systems allows us to better model complex adaptive systems. In the next module, we will expand on this foundation.